

End-to-End SOC Lab: Attack Simulation, Detection, Mitigation & SIEM Monitoring

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Environment:

Multi-Platform Log Ingestion (Kali Linux, Windows 10, Ubuntu)

Tools & Technologies Used:

VirtualBox (Lab Virtualization), OPNsense (Firewall & Logging), Nmap (Network Scanning), OpenVAS (Vulnerability Assessment), Hydra (Brute-Force Simulation), Enum4linux (SMB Enumeration), Sysmon (Windows Endpoint Monitoring), Splunk Cloud – accessed from Kali Linux (SIEM & Log Analysis), Windows Event Viewer, Linux Authentication Logs.

Phases Phases

◆ Phase 1: Visibility & Asset Discovery

- Objective: Establish network visibility and identify all active assets in the SOC lab environment.
- Activities: Verified the IP addressing scheme, configured the OPNsense router for WAN/LAN segmentation, and conducted Nmap scans to discover hosts and open services.
- Outcome: Created a baseline network topology identifying Kali, Ubuntu, and Windows hosts with their respective exposed services.

◆ Phase 2: Baseline Security and Logging

- Objective: Establish secure baseline configurations and enable logging for monitoring.
- Activities: Enabled firewall logging in OPNsense, configured static IPs, and implemented initial system hardening through service validation and firewall policies.
- Outcome: Developed a stable environment with logging enabled for traffic inspection and behavioral baseline creation.

◆ Phase 3: Detection & Vulnerability Assessment

- Objective: Identify vulnerabilities and observe suspicious activity using scanning tools.
- Activities: Conducted OpenVAS vulnerability scans on Ubuntu (192.168.1.70) and Windows (192.168.1.60), identifying security weaknesses across RPC and FTP services.
- Outcome: Documented system vulnerabilities and established high-fidelity visibility using OPNsense firewall logs.

◆ Phase 4: Incident Simulation & Response

- Objective: Simulate real-world attacks, apply remediation, and verify mitigation.
- Activities: Simulated brute-force attacks using Hydra and correlated findings with live firewall logs. Remediation included hardening exposed services and adjusting firewall policies to block unauthorized traffic.
- Outcome: Successfully verified vulnerability reduction through re-scanning, confirming that attack attempts no longer succeeded.

◆ Phase 5: Advanced Monitoring & SIEM Integration

- Objective: Enhance detection capabilities using centralized log management and real-time security analytics.
- Activities: Deployed Splunk Enterprise on Kali Linux and installed Universal Forwarders on endpoints. Manually configured inputs.conf and Event Log Reader permissions on Windows to bypass Home Edition restrictions.
- Outcome: Achieved a functional SOC monitoring infrastructure, successfully indexing 1,692 authentication events and 3,714 system events for real-time threat hunting.

Phase 1: Visibility & Asset Discovery.

Network Topology Diagram

Step 1: Verify IP Addresses

machine	IPs
kali	192.168.1.50
Ubuntu	192.168.1.70
Windows	192.168.1.60

Step 2: Network Discovery with Nmap

Nmap -sn 192.168.1.0/24

```
(mohammadkaif@mohammadkaif)-[~]
$ nmap -sn 192.168.1.0/24
Starting Nmap 7.95 ( https://nmap.org ) at 2026-01-14 11:23 IST
Nmap scan report for 192.168.1.1
Host is up (0.0030s latency).
MAC Address: 08:00:27:CF:07:DC (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
Nmap scan report for 192.168.1.60
Host is up (0.0011s latency).
MAC Address: 08:00:27:4E:B2:2C (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
Nmap scan report for 192.168.1.70
Host is up (0.00086s latency).
MAC Address: 08:00:27:70:0E:3D (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
Nmap scan report for 192.168.1.50
Host is up.
Nmap done: 256 IP addresses (4 hosts up) scanned in 2.16 seconds
```

IP Address Inventory

Machine	Role	Ip address	OS versio	MAC address
OPNsense	Firewall/Gateway	192.168.1.1	OPNsense	08:00:27:cf:07:dc
Kali	Attacker / Analyst	192.168.1.50	Kali Linux	08:00:27:f7:34:d1
Ubuntu	Web Server Target	192.168.1.70	Ubuntu Server	08:00:27:70:0e:3d
Windows	User Endpoint	192.168.1.60	Windows 10	08:00:27:4e:b2:2c

Asset Visibility Evidence

A. ARP Table (OPNsense)

Interfaces: Diagnostics: ARP Table

IP	MAC	Manufacturer	Interface	Interface name	Hostname
☐ 10.0.2.15	08:00:27:bc:9a:88	PCS Systemtechnik GmbH	em0	WAN	
☐ 10.0.2.2	52:54:00:12:35:02		em0	WAN	
☐ 192.168.1.1	08:00:27:cf:07:dc	PCS Systemtechnik GmbH	em1	LAN	
☐ 192.168.1.70	08:00:27:70:0e:3d	PCS Systemtechnik GmbH	em1	LAN	
☐ 192.168.1.60	08:00:27:4e:b2:2c	PCS Systemtechnik GmbH	em1	LAN	
☐ 192.168.1.50	08:00:27:f7:34:d1	PCS Systemtechnik GmbH	em1	LAN	

Showing 1 to 6 of 6 entries

Shows:

- IP ↔ MAC mapping
- LAN visibility

B. Firewall Live View (OPNsense)

OPNsense® Securing networks made easy root@OPNsense.internal

System
Interfaces
Firewall
Aliases
Automation
Categories
Groups
NAT
Rules
Shaper
Settings
Log Files
General
Live View
Overview
Plain View
Diagnostics
VPN
Services
Power

Firewall: Log Files: Live View

action contains pass + >> Choose template

☐ Select any of given criteria (or)

☒ Auto refresh
☐ Lookup hostnames
25

Interface	Time	Source	Destination	Proto	Label
▶ LAN →	2026-01-14T07:10:08	192.168.1.70:52074	192.168.1.1:443	tcp	anti-lockout rule ⓘ
▶ WAN ←	2026-01-14T07:09:44	10.0.2.15:123	86.107.100.245:123	udp	let out anything from firewall host itself (force gw) ⓘ
▶ WAN ←	2026-01-14T07:09:00	10.0.2.15:21087	151.101.1.91:443	tcp	let out anything from firewall host itself (force gw) ⓘ
▶ WAN ←	2026-01-14T07:09:00	10.0.2.15:13406	8.8.8.8:53	udp	let out anything from firewall host itself (force gw) ⓘ
▶ WAN ←	2026-01-14T07:09:00	10.0.2.15:59445	8.8.8.8:53	udp	let out anything from firewall host itself (force gw) ⓘ
▶ LAN →	2026-01-14T07:08:56	192.168.1.70:60084	192.168.1.1:443	tcp	anti-lockout rule ⓘ
▶ WAN ←	2026-01-14T07:08:39	10.0.2.15:123	86.107.100.245:123	udp	let out anything from firewall host itself (force gw) ⓘ
▶ WAN ←	2026-01-14T07:07:56	10.0.2.15:26399	151.101.1.91:443	tcp	let out anything from firewall host itself (force gw) ⓘ
▶ WAN ←	2026-01-14T07:07:56	10.0.2.15:49751	8.8.8.8:53	udp	let out anything from firewall host itself (force gw) ⓘ
▶ LAN →	2026-01-14T07:07:55	192.168.1.50:44980	192.168.1.1:443	tcp	anti-lockout rule ⓘ
▶ WAN ←	2026-01-14T07:07:55	10.0.2.15:47592	34.36.54.80:443	tcp	let out anything from firewall host itself (force gw) ⓘ
▶ WAN ←	2026-01-14T07:07:55	10.0.2.15:7495	8.8.8.8:53	udp	let out anything from firewall host itself (force gw) ⓘ
▶ WAN ←	2026-01-14T07:07:55	10.0.2.15:41135	8.8.8.8:53	udp	let out anything from firewall host itself (force gw) ⓘ
▶ WAN ←	2026-01-14T07:07:55	10.0.2.15:62629	8.8.8.8:53	udp	let out anything from firewall host itself (force gw) ⓘ

Shows:

- Source IPs
- Destination IPs
- Protocols
- Interface (WAN/LAN)

C. DHCP Leases



Services: ISC DHCPv4: Leases

☐ Show inactive

Search

All Interfaces

7

Interface	IP Address	MAC Address	Hostname	Description	Start	End	Status	State	Lease Type
No results found									

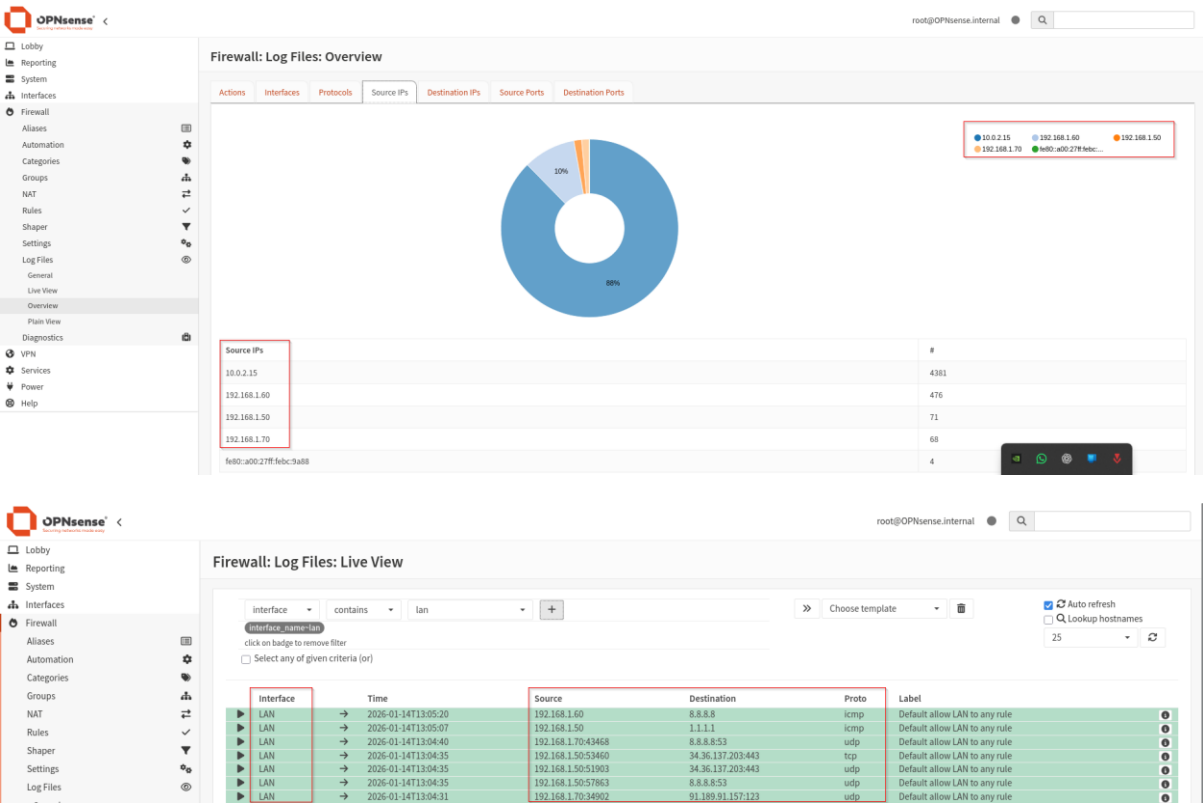
Showing 0 to 0 of 0 entries

Endpoints were configured manually using static IP addressing. As a result, DHCP lease records were not populated. Asset visibility was instead confirmed using the ARP table and firewall traffic logs, which provided accurate real-time endpoint identification

Phase 2: Baseline Security & Logging.

Network visibility (OPNsense)

During this phase, firewall logging was enabled and verified using OPNsense. Internal LAN traffic, including ICMP communication to external destinations, was successfully captured in live logs. The IP addressing scheme and firewall rules were validated to ensure proper segmentation and secure baseline configurations. This established normal traffic patterns and ensured visibility before proceeding to vulnerability assessment and attack simulations.



Verified firewall logging rules by capturing ICMP traffic from internal assets to external destinations.

The Sysmon (Windows)

A connectivity test was performed from the Windows system using ping 8.8.8.8. The successful ICMP replies (0% packet loss) confirmed external network access and proper LAN-to-WAN routing. This verified that firewall rules were functioning correctly and that traffic logging could capture outbound communication for monitoring and baseline analysis.

```

C:\Users\mohammadkaif\Downloads\Sysmon>ping 8.8.8.8

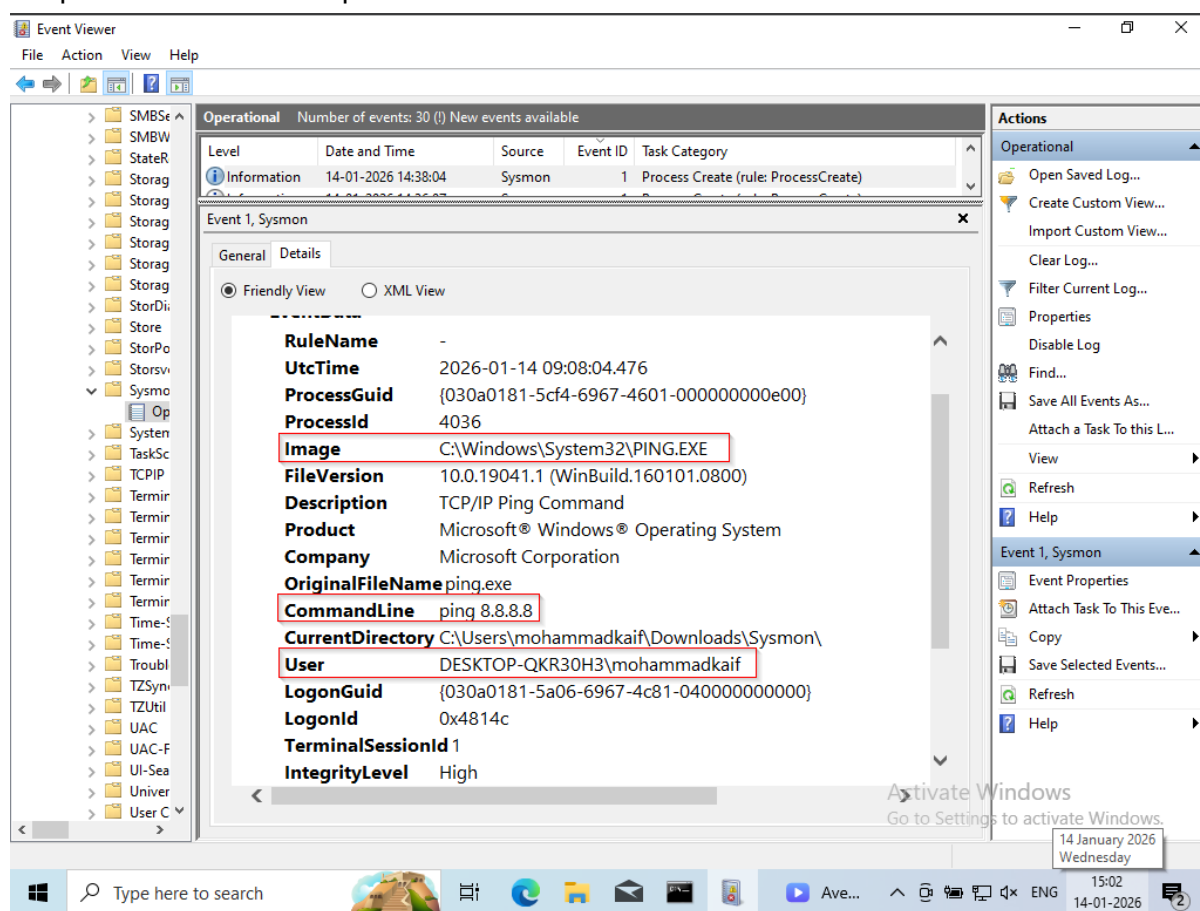
Pinging 8.8.8.8 with 32 bytes of data:
Reply from 8.8.8.8: bytes=32 time=5ms TTL=116
Reply from 8.8.8.8: bytes=32 time=6ms TTL=116
Reply from 8.8.8.8: bytes=32 time=5ms TTL=116
Reply from 8.8.8.8: bytes=32 time=7ms TTL=116

Ping statistics for 8.8.8.8:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 5ms, Maximum = 7ms, Average = 5ms

```

Sysmon Connectivity Verification

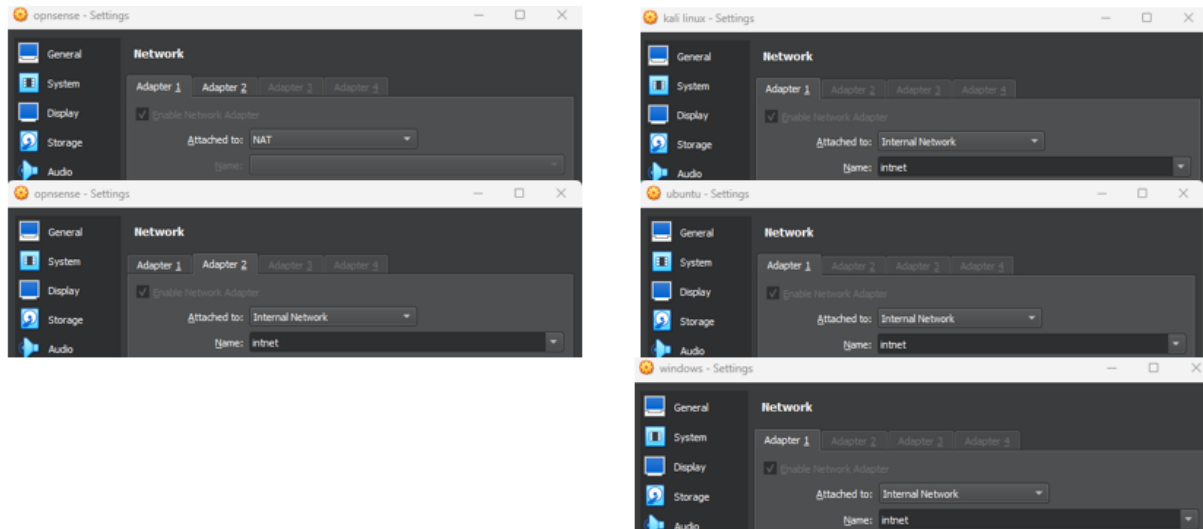
Sysmon was configured on the Windows system to monitor process creation events (Event ID 1). The logs captured execution of PING.EXE with the command line argument ping 8.8.8.8, including details such as user account, file path, and timestamp. This verified that Sysmon successfully records command-line activity, enabling detection of suspicious or abnormal process execution within the environment.



Implemented Sysmon ID 1 (Process Creation) to detect command-line execution anomalies.

Architecture configuration

The lab environment was configured using VirtualBox with an isolated **Internal Network (intnet)** to simulate a corporate LAN. Kali, Ubuntu, Windows, and OPNsense were connected to the same internal network segment, while OPNsense was configured with dual interfaces (WAN for external access and LAN for internal routing). This setup ensured proper network segmentation, controlled traffic flow, and realistic enterprise-style architecture for security monitoring and attack simulation.



Configured isolated internal network segments to simulate a corporate LAN environment

Phase 3: Vulnerability Scanning & Enumeration.

Ubuntu

- Running a Service scan on Ubuntu

```
(root@mohammadkaif)-[/home/mohammadkaif/Desktop]
# nmap -sV 192.168.1.70
Starting Nmap 7.98 ( https://nmap.org ) at 2026-01-29 11:07 +0530
Nmap scan report for 192.168.1.70
Host is up (0.0014s latency).
Not shown: 998 closed tcp ports (reset)
PORT      STATE SERVICE VERSION
21/tcp    open  ftp      vsftpd 3.0.5
80/tcp    open  http     Apache httpd 2.4.58 ((Ubuntu))
MAC Address: 08:00:27:70:0E:3D (Oracle VirtualBox virtual NIC)
Service Info: OS: Unix

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 7.04 seconds
```

Port : 80

Service: http

Version : Apache httpd 2.4.58

Port: 21

Service: ftp

Version: vsftpd 3.0.5

- Running vulnerability scan using nmap scripting engine, this command will throw lot of scripts at the target.

```
(root@mohammadkaif)-[/home/mohammadkaif/Desktop]
# nmap --script vuln 192.168.1.70
Starting Nmap 7.98 ( https://nmap.org ) at 2026-01-29 11:09 +0530
Nmap scan report for 192.168.1.70
Host is up (0.0055s latency).
Not shown: 998 closed tcp ports (reset)
PORT      STATE SERVICE
21/tcp    open  ftp
80/tcp    open  http
|_http-csrf: Couldn't find any CSRF vulnerabilities.
|_http-dombased-xss: Couldn't find any DOM based XSS.
|_http-stored-xss: Couldn't find any stored XSS vulnerabilities.
MAC Address: 08:00:27:70:0E:3D (Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 31.72 seconds
```

http-dombased-xss: Couldn't find any DOM based XSS.

http-stored-xss: Couldn't find any stored XSS.

http-csrf: Couldn't find any CSRF vulnerabilities.

Windows

- Running a Service scan on Windows

```
(mohammadkaif@mohammadkaif)-[~/Desktop]
$ nmap -sV 192.168.1.60
Starting Nmap 7.98 ( https://nmap.org ) at 2026-02-22 12:39 +0530
Nmap scan report for 192.168.1.60
Host is up (0.010s latency).
Not shown: 996 closed tcp ports (reset)
PORT      STATE SERVICE        VERSION
135/tcp    open  msrpc           Microsoft Windows RPC
139/tcp    open  netbios-ssn     Microsoft Windows netbios-ssn
445/tcp    open  microsoft-ds?
5357/tcp   open  http            Microsoft HTTPAPI httpd 2.0 (SSDP/UPnP)
MAC Address: 08:00:27:4E:B2:2C (Oracle VirtualBox virtual NIC)
Service Info: OS: Windows; CPE: cpe:/o:microsoft:windows

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 13.80 seconds
```

135/tcp – MSRPC (Microsoft Windows RPC)

139/tcp – NetBIOS-SSN (Microsoft Windows NetBIOS Session Service)

445/tcp – Microsoft-DS (SMB)

5357/tcp – HTTP (Microsoft HTTPAPI httpd 2.0 – SSDP/UPnP)

- Running vulnerability scan using nmap scripting engine, this command will throw lot of scripts at the target.

```
(mohammadkaif@mohammadkaif)-[~/Desktop]
$ nmap --script vuln 192.168.1.60
Starting Nmap 7.98 ( https://nmap.org ) at 2026-02-22 12:44 +0530
Nmap scan report for 192.168.1.60
Host is up (0.0034s latency).
Not shown: 996 closed tcp ports (reset)
PORT      STATE SERVICE
135/tcp    open  msrpc
139/tcp    open  netbios-ssn
445/tcp    open  microsoft-ds
5357/tcp   open  wsddapi
MAC Address: 08:00:27:4E:B2:2C (Oracle VirtualBox virtual NIC)

Host script results:
|_smb-vuln-ms10-061: Could not negotiate a connection:SMB: Failed to receive bytes: ERROR
|_samba-vuln-cve-2012-1182: Could not negotiate a connection:SMB: Failed to receive bytes: ERROR
|_smb-vuln-ms10-054: false

Nmap done: 1 IP address (1 host up) scanned in 26.92 seconds
```

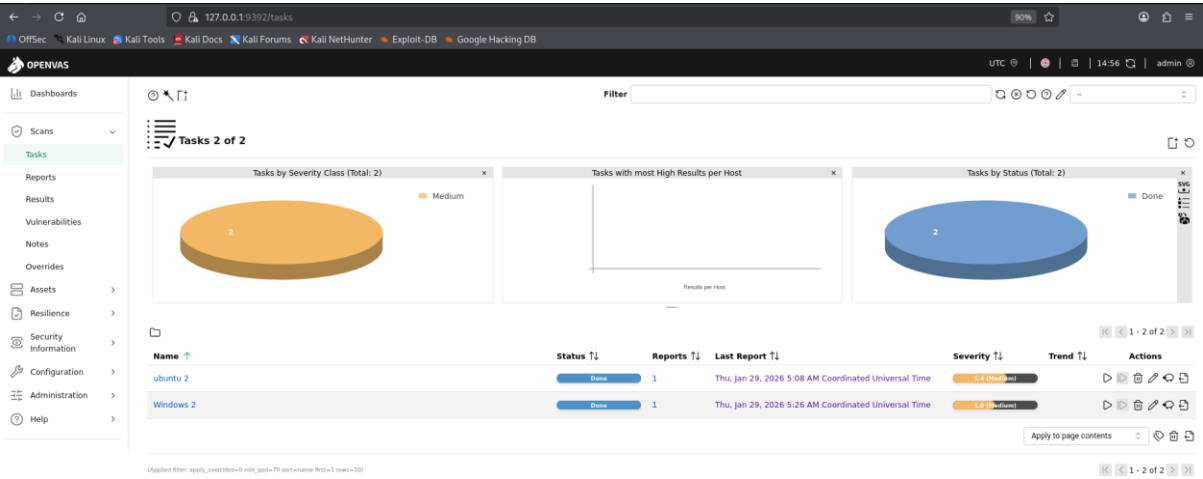
smb-vuln-ms10-061 → Connection failed (SMB negotiation error)

samba-vuln-cve-2012-1182 → Connection failed (SMB negotiation error)

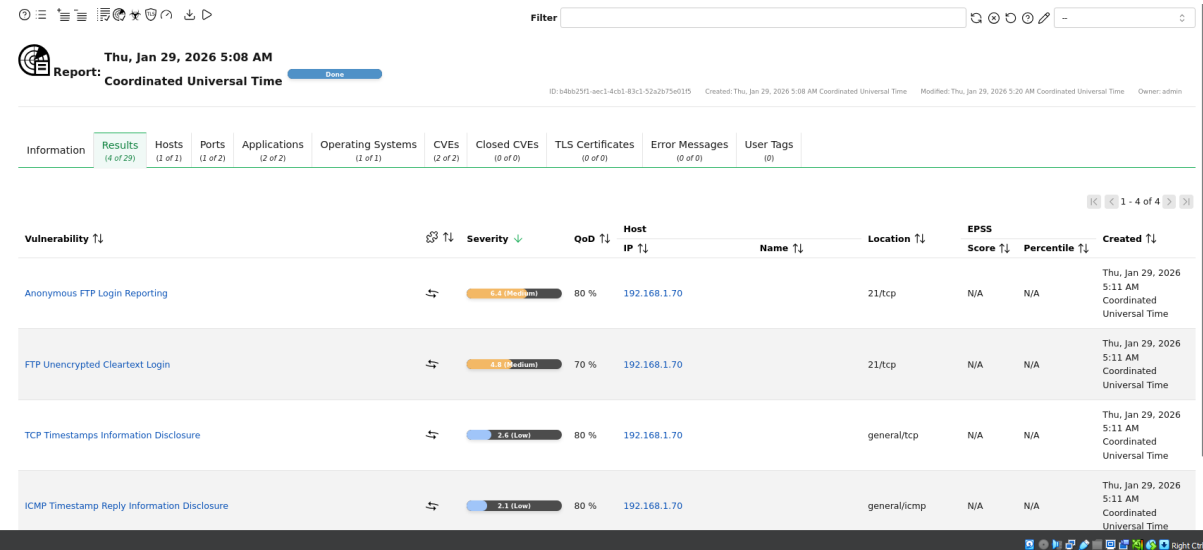
smb-vuln-ms10-054 → False (Not vulnerable)

OpenVas Scans

Vulnerability assessments were conducted using **OpenVAS** on the Ubuntu and Windows systems within the lab network. The scan results identified medium-severity vulnerabilities on both hosts, confirming the presence of misconfigurations and exposed services. The findings were documented and used to guide remediation efforts, forming the basis for further hardening and validation in subsequent phases.



Ubuntu



1. Anonymous FTP Login Reporting

Severity: Medium (6.4)

Port: 21/tcp

Meaning :

Ubuntu machine is running an FTP service that allows **anonymous login** (no username/password).

Why this matters

- Anyone on the network can access the FTP service

- Can be used for:
 - Unauthorized file access
 - Data staging
 - Malware drop locations

Anonymous FTP access increases the attack surface and allows unauthenticated users to interact with the system

2. FTP Unencrypted Cleartext Login

Severity: Medium (4.8)

Port: 21/tcp

Meaning:

FTP credentials (if used) are transmitted in **plain text**.

Why this matters

- Credentials can be sniffed
- Enables credential reuse attacks
- Fails basic security hardening standards

Use of unencrypted FTP exposes credentials to interception and violates secure communication best practices.

3. TCP Timestamps Information Disclosure

Severity: Low (2.6)

Scope: general/tcp

Meaning:

The system leaks **TCP timestamp values**, which can help attackers:

- Estimate system uptime
- Fingerprint OS behavior

This finding represents low-risk information disclosure that could assist advanced attackers during reconnaissance.

4. ICMP Timestamp Reply Information Disclosure

Severity: Low (2.1)

Scope: general/icmp

Meaning:

The host replies to ICMP timestamp requests, revealing time-related system info.

ICMP timestamp responses provide unnecessary information that can aid reconnaissance

Windows

Filter									
Report: Thu, Jan 29, 2026 5:26 AM Coordinated Universal Time									
ID: 87c499c-7591-499f-9476-6657a8fc959 Created: Thu, Jan 29, 2026 5:26 AM Coordinated Universal Time Modified: Thu, Jan 29, 2026 6:59 AM Coordinated Universal Time Owner: admin									
Information	Results (2 of 17)	Hosts (1 of 1)	Ports (1 of 5)	Applications (0 of 0)	Operating Systems (1 of 1)	CVEs (1 of 1)	Closed CVEs (0 of 0)	TLS Certificates (0 of 0)	Error Messages (5 of 5)
Vulnerability Severity QoD Host IP Name Location EPSS Score Percentile Created									
DCE/RPC and MSRPC Services Enumeration Reporting 5.0 (Medium) 80 % 192.168.1.60 135/tcp N/A N/A Thu, Jan 29, 2026 6:03 AM Coordinated Universal Time									
ICMP Timestamp Reply Information Disclosure 2.1 (Low) 80 % 192.168.1.60 general/cmp N/A N/A Thu, Jan 29, 2026 5:42 AM Coordinated Universal Time									
(Applied filter: apply_overrides=0 level=ctrl rows=100 min_qod=70 first=1, sort=reverse:severity)									

1. DCE/RPC and MSRPC Services Enumeration

Severity: Medium (5.0)

Port: 135/tcp

What it means (plain English):

Your Windows machine exposes **RPC services** that can be **enumerated remotely**.

Why this matters

- Port **135 (RPC Endpoint Mapper)** is a known reconnaissance target
- Allows attackers to:
 - Identify running services
 - Map Windows internals
 - Prepare for lateral movement (in enterprise networks)
 - This is **NOT an exploit**, it's **information exposure**.

The exposure of DCE/RPC services allows remote enumeration of Windows service interfaces, increasing the system's reconnaissance surface

2. ICMP Timestamp Reply Information Disclosure

Severity: Low (2.1)

Protocol: ICMP

What it means:

The Windows host responds to ICMP timestamp requests, revealing time-related system data.

Why this matters

- Helps OS fingerprinting
- Can assist advanced attackers during recon
- Low risk on its own

This represents low-risk information disclosure consistent with default Windows networking behavior.

Phase 4 Incident simulation

Attack and Detection

Ubuntu

Attack 1

During this phase, real-world attack scenarios were simulated against the Ubuntu system. A brute-force attack using Hydra targeted the SSH service, generating multiple failed login attempts. These attempts were successfully detected in `/var/log/auth.log`, confirming authentication monitoring was functioning correctly.

Additionally, Nmap scans identified exposed services including FTP (anonymous login enabled), SSH, and HTTP. The attack activity was correlated with system logs, demonstrating effective detection, analysis, and validation of suspicious behavior within the lab environment.

```
mohammadkaif@mohammadkaif: ~/Desktop
$ hydra -l root -P /usr/share/wordlists/rockyou.txt ssh://192.168.1.70
Hydra v9.6 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or for illegal purposes (this is non-binding, these ** ignore laws and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-02-21 13:01:44
[WARNING] Many SSH configurations limit the number of parallel tasks, it is recommended to reduce the tasks: use -t 4
[WARNING] Restorefile (you have 10 seconds to abort... (use option -i to skip waiting)) from a previous session found, to prevent overwriting, ./hydra.restore
[DATA] max 16 tasks per 1 server, overall 16 tasks, 14344399 login tries (l:1/p:14344399), ~896525 tries per task
[DATA] attacking ssh://192.168.1.70:22/

mohammadkaif@mohammadkaif-ubuntu: ~$ sudo tail -f /var/log/auth.log
2026-01-31T12:59:55.423880+05:30 mohammadkaif-ubuntu sshd[7117]: Failed password for root from 192.168.1.50 port 34122 ssh2
2026-01-31T12:59:55.486553+05:30 mohammadkaif-ubuntu sshd[7130]: Failed password for root from 192.168.1.50 port 49310 ssh2
2026-01-31T12:59:55.520148+05:30 mohammadkaif-ubuntu sshd[7119]: Failed password for root from 192.168.1.50 port 34128 ssh2
2026-01-31T12:59:55.686044+05:30 mohammadkaif-ubuntu sshd[7123]: Failed password for root from 192.168.1.50 port 49260 ssh2
2026-01-31T12:59:56.426161+05:30 mohammadkaif-ubuntu sshd[7126]: Failed password for root from 192.168.1.50 port 49274 ssh2
2026-01-31T12:59:57.078030+05:30 mohammadkaif-ubuntu sshd[7132]: Failed password for root from 192.168.1.50 port 49324 ssh2
2026-01-31T12:59:57.083456+05:30 mohammadkaif-ubuntu sshd[7128]: Failed password for root from 192.168.1.50 port 49286 ssh2
2026-01-31T12:59:57.083768+05:30 mohammadkaif-ubuntu sshd[7113]: Failed password for root from 192.168.1.50 port 34114 ssh2
2026-01-31T12:59:57.083920+05:30 mohammadkaif-ubuntu sshd[7121]: Failed password for root from 192.168.1.50 port 34132 ssh2
2026-01-31T12:59:57.175752+05:30 mohammadkaif-ubuntu sshd[7115]: Failed password for root from 192.168.1.50 port 34116 ssh2
2026-01-31T12:59:57.381376+05:30 mohammadkaif-ubuntu sshd[7111]: Failed password for root from 192.168.1.50 port 34106 ssh2
2026-01-31T12:59:57.411316+05:30 mohammadkaif-ubuntu sshd[7109]: Failed password for root from 192.168.1.50 port 34104 ssh2
2026-01-31T12:59:57.725564+05:30 mohammadkaif-ubuntu sshd[7130]: Failed password for root from 192.168.1.50 port 49310 ssh2
2026-01-31T12:59:57.931633+05:30 mohammadkaif-ubuntu sshd[7117]: Failed password for root from 192.168.1.50 port 34122 ssh2
2026-01-31T12:59:57.932238+05:30 mohammadkaif-ubuntu sshd[7123]: Failed password for root from 192.168.1.50 port 49260 ssh2
```



```

(mohammadkaif@mohammadkaif)-[~/Desktop]
$ nmap -A -T4 -p- 192.168.1.70
Starting Nmap 7.98 ( https://nmap.org ) at 2026-02-21 13:06 +0530
Nmap scan report for 192.168.1.70
Host is up (0.0010s latency).
Not shown: 65532 closed tcp ports (reset)
PORT      STATE SERVICE VERSION
21/tcp    open  ftp      vsftpd 3.0.5
|_ftp-anon: Anonymous FTP login allowed (FTP code 230)
|_ftp-syst:
|  STAT:
|  FTP server status:
|    Connected to ::ffff:192.168.1.50
|    Logged in as ftp
|    TYPE: ASCII
|    No session bandwidth limit
|    Session timeout in seconds is 300
|    Control connection is plain text
|    Data connections will be plain text
|    At session startup, client count was 3
|    vsFTPD 3.0.5 - secure, fast, stable
|_End of status
22/tcp    open  ssh      OpenSSH 9.6p1 Ubuntu 3ubuntu13.14 (Ubuntu Linux; protocol 2.0)
|_ssh-hostkey:
|  256 58:f3:9b:24:84:1d:da:d6:f5:8d:38:c4:30:65:8f:13 (ECDSA)
|_ 256 6e:34:30:eb:7c:b7:09:70:e5:5e:1f:4b:9f:97:af:87 (ED25519)
80/tcp    open  http      Apache httpd 2.4.58 ((Ubuntu))
|_http-title: Apache2 Ubuntu Default Page: It works
|_http-server-header: Apache/2.4.58 (Ubuntu)
MAC Address: 08:00:27:70:0E:3D (Oracle VirtualBox virtual NIC)
Device type: general purpose
Running: Linux 4.X|5.X
OS CPE: cpe:/o:linux:linux_kernel:4 cpe:/o:linux:linux_kernel:5
OS details: Linux 4.15 - 5.19, OpenWrt 21.02 (Linux 5.4)
Network Distance: 1 hop
Service Info: OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

TRACEROUTE
HOP RTT      ADDRESS
1   1.03 ms  192.168.1.70

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 34.74 seconds

```

```

mohammadkaif@mohammadkaif-ubuntu:~$ sudo tail -f /var/log/auth.log
2026-02-21T13:04:41.753697+05:30 mohammadkaif-ubuntu sshd[4646]: PAM 5 more authentication failures; logname=uid=0 euid=0 tty=ssh ruser= rhost=192.168.1.50 user=root
2026-02-21T13:04:41.753819+05:30 mohammadkaif-ubuntu sshd[4646]: PAM service(sshd) ignoring max retries; 6 > 3
2026-02-21T13:04:41.760600+05:30 mohammadkaif-ubuntu sshd[4655]: Connection closed by authenticating user root 192.168.1.50 port 48246 [preauth]
2026-02-21T13:04:41.760738+05:30 mohammadkaif-ubuntu sshd[4655]: PAM 2 more authentication failures; logname=uid=0 euid=0 tty=ssh ruser= rhost=192.168.1.50 user=root
2026-02-21T13:04:41.760773+05:30 mohammadkaif-ubuntu sshd[4663]: Connection closed by authenticating user root 192.168.1.50 port 52528 [preauth]
2026-02-21T13:05:01.533521+05:30 mohammadkaif-ubuntu CRON[4667]: pam_unix(cron:session): session opened for user root(uid=0) by root(uid=0)
2026-02-21T13:05:01.543930+05:30 mohammadkaif-ubuntu CRON[4667]: pam_unix(cron:session): session closed for user root
2026-02-21T13:05:04.674847+05:30 mohammadkaif-ubuntu sudo: pam_unix(sudo:session): session closed for user root
2026-02-21T13:05:06.491517+05:30 mohammadkaif-ubuntu sudo: mohammadkaif : TTY=pts/0 ; PWD=/home/mohammadkaif ; USER=root ; COMMAND=/usr/bin/tail -f /var/log/auth.log
2026-02-21T13:05:06.492219+05:30 mohammadkaif-ubuntu sudo: pam_unix(sudo:session): session opened for user root(uid=0) by mohammadkaif(uid=0)
2026-02-21T13:05:07.916755+05:30 mohammadkaif-ubuntu sshd[1080]: exited MaxStartups throttling after 00:03:12, 54 connections dropped
2026-02-21T13:05:07.935558+05:30 mohammadkaif-ubuntu sshd[4674]: Connection closed by 192.168.1.50 port 39142
2026-02-21T13:05:15.293804+05:30 mohammadkaif-ubuntu sshd[4680]: error: Protocol major versions differ: 2 vs. 1
2026-02-21T13:05:15.294996+05:30 mohammadkaif-ubuntu sshd[4680]: banner exchange: Connection from 192.168.1.50 port 39144: could not read protocol version
2026-02-21T13:05:15.295109+05:30 mohammadkaif-ubuntu sshd[4684]: error: Protocol major versions differ: 2 vs. 1
2026-02-21T13:05:15.295127+05:30 mohammadkaif-ubuntu sshd[4684]: banner exchange: Connection from 192.168.1.50 port 39146: could not read protocol version
2026-02-21T13:05:15.599728+05:30 mohammadkaif-ubuntu sshd[4688]: Unable to negotiate with 192.168.1.50 port 39152: no matching host key type found. Their offer: ssh-dss [preauth]
2026-02-21T13:05:15.531731+05:30 mohammadkaif-ubuntu sshd[4694]: Unable to negotiate with 192.168.1.50 port 39168: no matching host key type found. Their offer: ssh-rsa [preauth]
2026-02-21T13:05:15.687486+05:30 mohammadkaif-ubuntu sshd[4696]: Connection closed by 192.168.1.50 port 39172 [preauth]
2026-02-21T13:05:15.624306+05:30 mohammadkaif-ubuntu sshd[4698]: Unable to negotiate with 192.168.1.50 port 39186: no matching host key type found. Their offer: ecdsa-sha2-nistp384 [preauth]
2026-02-21T13:05:15.650482+05:30 mohammadkaif-ubuntu sshd[4700]: Unable to negotiate with 192.168.1.50 port 39188: no matching host key type found. Their offer: ecdsa-sha2-nistp521 [preauth]
2026-02-21T13:05:15.677486+05:30 mohammadkaif-ubuntu sshd[4702]: Connection closed by 192.168.1.50 port 39192 [preauth]
2026-02-21T13:06:39.676738+05:30 mohammadkaif-ubuntu sshd[4748]: Connection closed by 192.168.1.50 port 53976
2026-02-21T13:06:46.992612+05:30 mohammadkaif-ubuntu sshd[4752]: error: Protocol major versions differ: 2 vs. 1
2026-02-21T13:06:46.992765+05:30 mohammadkaif-ubuntu sshd[4752]: banner exchange: Connection from 192.168.1.50 port 53496: could not read protocol version
2026-02-21T13:06:46.993579+05:30 mohammadkaif-ubuntu sshd[4754]: error: Protocol major versions differ: 2 vs. 1
2026-02-21T13:06:46.993626+05:30 mohammadkaif-ubuntu sshd[4754]: banner exchange: Connection from 192.168.1.50 port 53500: could not read protocol version
2026-02-21T13:06:47.272556+05:30 mohammadkaif-ubuntu sshd[4760]: Unable to negotiate with 192.168.1.50 port 53510: no matching host key type found. Their offer: ssh-dss [preauth]
2026-02-21T13:06:47.297831+05:30 mohammadkaif-ubuntu sshd[4765]: Unable to negotiate with 192.168.1.50 port 53514: no matching host key type found. Their offer: ssh-rsa [preauth]
2026-02-21T13:06:47.365584+05:30 mohammadkaif-ubuntu sshd[4767]: Connection closed by 192.168.1.50 port 53526 [preauth]
2026-02-21T13:06:47.396804+05:30 mohammadkaif-ubuntu sshd[4769]: Unable to negotiate with 192.168.1.50 port 53540: no matching host key type found. Their offer: ecdsa-sha2-nistp384 [preauth]
2026-02-21T13:06:47.430889+05:30 mohammadkaif-ubuntu sshd[4771]: Unable to negotiate with 192.168.1.50 port 53546: no matching host key type found. Their offer: ecdsa-sha2-nistp521 [preauth]
2026-02-21T13:06:47.484469+05:30 mohammadkaif-ubuntu sshd[4773]: Connection closed by 192.168.1.50 port 53556 [preauth]

```

Attack 2 : Anonymous FTP Login

An anonymous FTP login was successfully performed against the Ubuntu system (192.168.1.70), confirming that the FTP service (vsFTPD 3.0.5) allowed unauthenticated access. The connection and login activity were verified in /var/log/vsftpd.log, which recorded successful anon login attempts from the attacking host (192.168.1.50).

This demonstrated a security misconfiguration, as anonymous FTP access can expose sensitive files and increase the attack surface if not properly restricted.

```
(mohammadkaif@mohammadkaif)-[~/Desktop]
$ ftp 192.168.1.70
Connected to 192.168.1.70.
220 (vsFTPD 3.0.5)
Name (192.168.1.70:mohammadkaif): anonymous
331 Please specify the password.
Password:
230 Login successful.
Remote system type is UNIX.
Using binary mode to transfer files.
ftp> ls
229 Entering Extended Passive Mode (|||8359|)
150 Here comes the directory listing.
226 Directory send OK.
ftp> exit
221 Goodbye.
```

```
mohammadkaif@mohammadkaif-ubuntu:~$ sudo cat /var/log/vsftpd.log
Sat Feb 21 13:05:07 2026 [pid 4676] CONNECT: Client "::ffff:192.168.1.50"
Sat Feb 21 13:05:15 2026 [pid 4681] CONNECT: Client "::ffff:192.168.1.50"
Sat Feb 21 13:05:15 2026 [pid 4682] CONNECT: Client "::ffff:192.168.1.50"
Sat Feb 21 13:05:15 2026 [pid 4686] CONNECT: Client "::ffff:192.168.1.50"
Sat Feb 21 13:05:15 2026 [pid 4687] CONNECT: Client "::ffff:192.168.1.50"
Sat Feb 21 13:05:15 2026 [pid 4679] [ftp] OK LOGIN: Client "::ffff:192.168.1.50", anon password "IEUser@"
Sat Feb 21 13:05:15 2026 [pid 4683] [ftp] OK LOGIN: Client "::ffff:192.168.1.50", anon password "IEUser@"
Sat Feb 21 13:05:15 2026 [pid 4678] [ftp] OK LOGIN: Client "::ffff:192.168.1.50", anon password "IEUser@"
Sat Feb 21 13:05:15 2026 [pid 4705] CONNECT: Client "::ffff:192.168.1.50"
Sat Feb 21 13:05:15 2026 [pid 4707] CONNECT: Client "::ffff:192.168.1.50"
Sat Feb 21 13:05:15 2026 [pid 4709] CONNECT: Client "::ffff:192.168.1.50"
Sat Feb 21 13:05:15 2026 [pid 4711] CONNECT: Client "::ffff:192.168.1.50"
Sat Feb 21 13:05:15 2026 [pid 4713] CONNECT: Client "::ffff:192.168.1.50"
Sat Feb 21 13:05:15 2026 [pid 4715] CONNECT: Client "::ffff:192.168.1.50"
Sat Feb 21 13:06:39 2026 [pid 4749] CONNECT: Client "::ffff:192.168.1.50"
Sat Feb 21 13:06:46 2026 [pid 4751] CONNECT: Client "::ffff:192.168.1.50"
Sat Feb 21 13:06:46 2026 [pid 4755] CONNECT: Client "::ffff:192.168.1.50"
Sat Feb 21 13:06:46 2026 [pid 4758] CONNECT: Client "::ffff:192.168.1.50"
Sat Feb 21 13:06:46 2026 [pid 4759] CONNECT: Client "::ffff:192.168.1.50"
Sat Feb 21 13:06:47 2026 [pid 4753] [ftp] OK LOGIN: Client "::ffff:192.168.1.50", anon password "IEUser@"
Sat Feb 21 13:06:47 2026 [pid 4757] [ftp] OK LOGIN: Client "::ffff:192.168.1.50", anon password "IEUser@"
Sat Feb 21 13:06:47 2026 [pid 4756] [ftp] OK LOGIN: Client "::ffff:192.168.1.50", anon password "IEUser@"
Sat Feb 21 13:06:47 2026 [pid 4776] CONNECT: Client "::ffff:192.168.1.50"
Sat Feb 21 13:06:47 2026 [pid 4778] CONNECT: Client "::ffff:192.168.1.50"
Sat Feb 21 13:06:47 2026 [pid 4780] CONNECT: Client "::ffff:192.168.1.50"
Sat Feb 21 13:06:47 2026 [pid 4782] CONNECT: Client "::ffff:192.168.1.50"
Sat Feb 21 13:06:47 2026 [pid 4784] CONNECT: Client "::ffff:192.168.1.50"
Sat Feb 21 13:06:47 2026 [pid 4786] CONNECT: Client "::ffff:192.168.1.50"
Sat Feb 21 13:08:34 2026 [pid 4788] CONNECT: Client "::ffff:192.168.1.50"
Sat Feb 21 13:08:52 2026 [pid 4787] [ftp] OK LOGIN: Client "::ffff:192.168.1.50", anon password "?"
```


Windows

Attack 1: SMB Enumeration

SMB enumeration was performed against the Windows system (192.168.1.60) using enum4linux. The scan identified the system as part of the WORKGROUP domain and revealed active SMB services, including File Server and Workstation services. Known usernames such as administrator and guest were also identified.

Although anonymous session access was not permitted, the enumeration confirmed that SMB services were exposed and accessible, increasing the potential attack surface for further exploitation attempts.

```
(mohammadkaif@mohammadkaif)-[~/Desktop]
$ enum4linux -a 192.168.1.60
Starting enum4linux v0.9.1 ( http://labs.portcullis.co.uk/application/enum4linux/ ) on Sat Feb 21 13:16:30 2026

===== ( Target Information ) =====
Target ..... 192.168.1.60
RID Range ..... 500-550,1000-1050
Username ..... ''
Password ..... ''
Known Usernames .. administrator, guest, krbtgt, domain admins, root, bin, none

===== ( Enumerating Workgroup/Domain on 192.168.1.60 ) =====

[+] Got domain/workgroup name: WORKGROUP

===== ( Nbtstat Information for 192.168.1.60 ) =====

Looking up status of 192.168.1.60
DESKTOP-QKR30H3 <00> - B <ACTIVE> Workstation Service
WORKGROUP <00> - <GROUP> B <ACTIVE> Domain/Workgroup Name
DESKTOP-QKR30H3 <20> - B <ACTIVE> File Server Service
WORKGROUP <1e> - <GROUP> B <ACTIVE> Browser Service Elections
WORKGROUP <1d> - B <ACTIVE> Master Browser
.._MSBROWSE_. <01> - <GROUP> B <ACTIVE> Master Browser

MAC Address = 08-00-27-4E-B2-2C

===== ( Session Check on 192.168.1.60 ) =====

[E] Server doesn't allow session using username '', password ''. Aborting remainder of tests.
```

Attack 2: SMB Brute Force

An SMB brute-force attack was conducted against the Windows system (192.168.1.60) targeting the administrator account. Multiple login attempts were generated, resulting in repeated authentication failures.

Windows Event Viewer logs (Event ID 4625 – Failed Logon) confirmed the attack activity, showing multiple failed login attempts within a short time frame. The account was eventually locked out, demonstrating effective detection and built-in security controls against brute-force attacks.

The image displays a terminal window and a Windows Event Viewer window. The terminal window shows the execution of a crackmapexec smb command against 192.168.1.60, targeting the administrator account with a password list from /usr/share/wordlists/rockyou.txt. The output shows multiple failed login attempts for various passwords, including 123456, 12345, password, iloveyou, princess, 1234567, and rockyou, resulting in a STATUS_ACCOUNT_LOCKED_OUT error.

The Windows Event Viewer window shows the Security log with 5,920 events. A filtered view shows 17 events of type Audit Failure (Event ID 4625) from the source Microsoft Windows security auditing. The details of Event 4625 are shown, indicating an account failed to log on. The subject is Security ID: NULL SID. The log name is Security, source is Microsoft Windows security, event ID is 4625, level is Information, user is N/A, and computer is DESKTOP-QKR30H3.

Keywords	Date and Time	Source	Event ID	Task Category
Audit Failure	21-02-2026 13:39:34	Microsoft Windows security auditing.	4625	Logon
Audit Failure	21-02-2026 13:39:34	Microsoft Windows security auditing.	4625	Logon
Audit Failure	21-02-2026 13:39:34	Microsoft Windows security auditing.	4625	Logon
Audit Failure	21-02-2026 13:39:34	Microsoft Windows security auditing.	4625	Logon
Audit Failure	21-02-2026 13:39:34	Microsoft Windows security auditing.	4625	Logon
Audit Failure	21-02-2026 13:39:34	Microsoft Windows security auditing.	4625	Logon
Audit Failure	21-02-2026 13:39:33	Microsoft Windows security auditing.	4625	Logon
Audit Failure	21-02-2026 13:37:29	Microsoft Windows security auditing.	4625	Logon
Audit Failure	21-02-2026 13:36:32	Microsoft Windows security auditing.	4625	Logon

Event 4625, Microsoft Windows security auditing.

General Details

An account failed to log on.

Subject:

Security ID: NULL SID

Account Name:

Log Name: Security

Source: Microsoft Windows security

Event ID: 4625

Level: Information

User: N/A

OpCode: Info

More Information: [Event Log Online Help](#)

Logged: 21-02-2026 13:39:34

Task Category: Logon

Keywords: Audit Failure

Computer: DESKTOP-QKR30H3

Mitigation

Ubuntu

Step 1: Deploy Fail2Ban (Automated Defense)

Fail2Ban is an incredible tool that actively monitors your system logs for failed authentication attempts and dynamically updates your firewall to block the attacker's IP address.

1. Install the packages

```
mohammadkaif@mohammadkaif-ubuntu:~$ sudo apt update && sudo apt install fail2ban -y
```

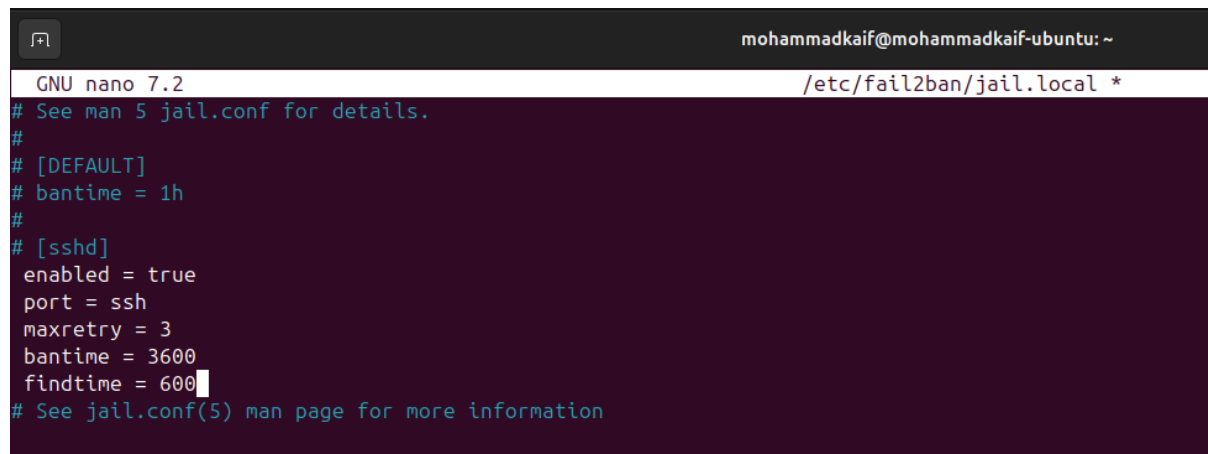
2. Create a local configuration file:

It is best practice to copy the default configuration rather than editing it directly so your changes are preserved during updates.

```
mohammadkaif@mohammadkaif-ubuntu:~$ sudo cp /etc/fail2ban/jail.conf /etc/fail2ban/jail.local  
mohammadkaif@mohammadkaif-ubuntu:~$
```

3. Configure the jails:

Open the file with `sudo nano /etc/fail2ban/jail.local`. Scroll down to find the `[sshd]` section and ensure it is enabled. Make sure your configurations look like this:



```
mohammadkaif@mohammadkaif-ubuntu: ~  
GNU nano 7.2 /etc/fail2ban/jail.local *  
# See man 5 jail.conf for details.  
#  
# [DEFAULT]  
# bantime = 1h  
#  
# [sshd]  
enabled = true  
port = ssh  
maxretry = 3  
bantime = 3600  
findtime = 600  
# See jail.conf(5) man page for more information
```

4. Enforce Rate Limiting with UFW

```
mohammadkaif@mohammadkaif-ubuntu:~$ sudo ufw limit ssh
```

If only your IT subnet (e.g., 192.168.1.0/24) needs access to the file shares, you can restrict access:

```
sudo ufw allow from 192.168.1.0/24 to any port 445
```

```
sudo ufw deny 445
```

Mitigation

Windows

The following remediation steps are recommended based on the identified SMB brute-force and enumeration findings:

- Restrict SMB access (ports 139 and 445) using Windows Defender Firewall to allow connections only from trusted internal hosts.
- Enforce strong password and account lockout policies to prevent brute-force attacks.
- Disable anonymous access and the Guest account to reduce SMB enumeration risks.
- Limit or disable unnecessary services such as File and Printer Sharing if not required.
- Enable advanced auditing and centralized logging (Security Logs and Sysmon) for better monitoring.
- Apply the latest Windows security updates and patches to address potential vulnerabilities.

Phase 5: Advanced Monitoring & SIEM Integration

Ubuntu

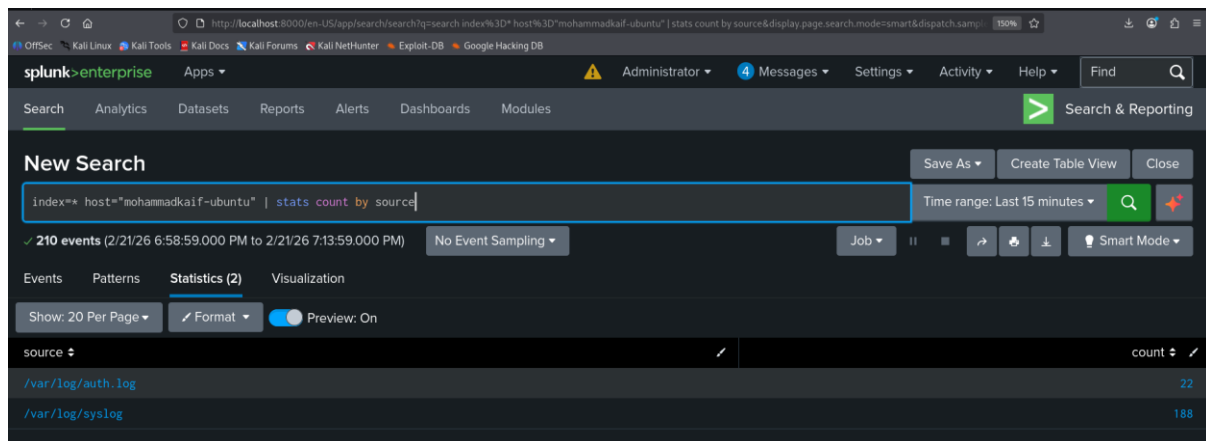
Document your configuration process to show the technical rigor of your setup:

- **Service Initialization:**
 - `cd /opt/splunkforwarder/bin`
 - `sudo ./splunk start --accept-license` (Note: This established the local administrative account).
 - `sudo ./splunk enable boot-start` (Note: This ensures the "spy" remains active even after a system reboot).
- **Network Bridging:**
 - `sudo ./splunk add forward-server [KALI_IP_ADDRESS]:9997.`
 - *Report Tip:* Include a screenshot of `sudo ./splunk list forward-server` showing the status as **Active**.
- **Security Log Ingestion:**
 - **Auth Monitoring:** `sudo ./splunk add monitor /var/log/auth.log` (Targets: SSH Brute-force and sudo abuse).
 - **System Monitoring:** `sudo ./splunk add monitor /var/log/syslog` (Targets: Service failures and kernel alerts).
 - **Activity Tracking:** `sudo ./splunk add monitor /home/./.bash_history` (Targets: Capturing the attacker's actual typed commands).

Evidence Query: Show a table of sources using: index=* host="mohammadkaif-ubuntu" | stats count by source.

Log Breakdown:

- **auth.log:** Point to the 1,692 events in your screenshot as evidence of tracking login attempts.
- **syslog:** Highlight the 3,714 events as a baseline for system health monitoring.



New Search

index=* host="mohammadkaif-ubuntu" | stats count by source

Time range: Last 15 minutes

210 events (2/21/26 6:58:59.000 PM to 2/21/26 7:13:59.000 PM) No Event Sampling

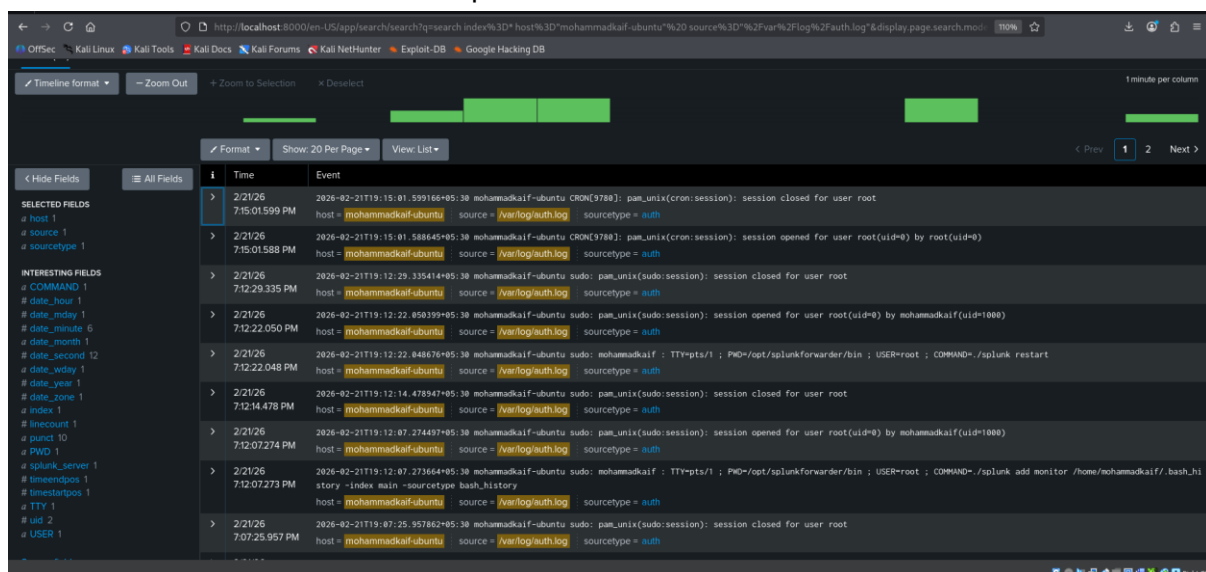
Events Patterns Statistics (2) Visualization

Show: 20 Per Page Format Preview: On

source	count
/var/log/auth.log	22
/var/log/syslog	188

1. The "Auth.log" Section (Identity & Access)

- **What it tracks:** All authentication events, including SSH logins, sudo attempts, and new user creations.
- **Security Value:** This is your primary defense against **Brute-Force attacks**. You can point to the 1,692 events in your screenshot as a successful "baseline" of activity.
- **Report Highlight:** Note that this log captures the **Source IP** of attackers, which is essential for incident response.



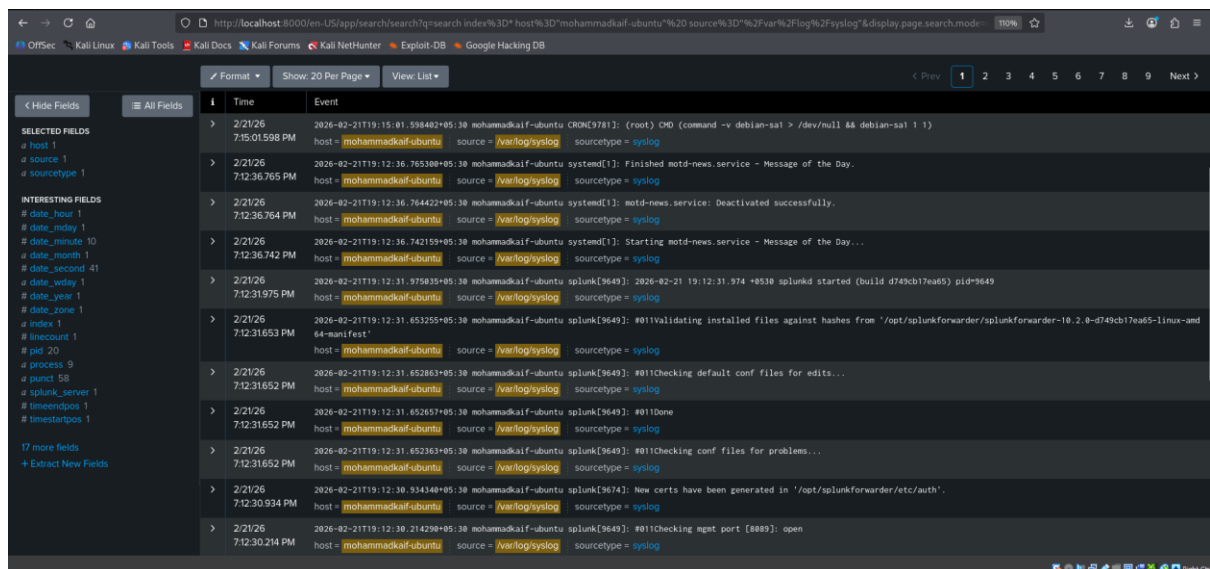
Timeline format Zoom Out Zoom to Selection Deselect

Format Show: 20 Per Page View: List

Time	Event
2/21/26 7:15:01:599 PM	2026-02-21T19:15:01.599166+05:30 mohammadkaif-ubuntu CRON(3788): pam_unix(cron:session): session closed for user root
2/21/26 7:15:01:588 PM	2026-02-21T19:15:01.588645+05:30 mohammadkaif-ubuntu CRON(3788): pam_unix(cron:session): session opened for user root(uid=0) by root(uid=0)
2/21/26 7:12:22:050 PM	2026-02-21T19:12:22.050399+05:30 mohammadkaif-ubuntu sudo: pam_unix(sudo:session): session opened for user root(uid=0) by mohammadkaif(uid=1000)
2/21/26 7:12:22:048 PM	2026-02-21T19:12:22.048676+05:30 mohammadkaif-ubuntu sudo: mohammadkaif : TTY=pts/1 ; PWD=/opt/splunkforwarder/bin ; USER=root ; COMMAND=/splunk restart
2/21/26 7:12:14:478 PM	2026-02-21T19:12:14.478947+05:30 mohammadkaif-ubuntu sudo: pam_unix(sudo:session): session closed for user root
2/21/26 7:12:07:274 PM	2026-02-21T19:12:07.274457+05:30 mohammadkaif-ubuntu sudo: pam_unix(sudo:session): session opened for user root(uid=0) by mohammadkaif(uid=1000)
2/21/26 7:12:07:273 PM	2026-02-21T19:12:07.273664+05:30 mohammadkaif-ubuntu sudo: mohammadkaif : TTY=pts/1 ; PWD=/opt/splunkforwarder/bin ; USER=root ; COMMAND=/splunk add monitor /home/mohammadkaif/.bash_history -index main -sourcetype bash_history
2/21/26 7:07:25:957 PM	2026-02-21T19:07:25.957862+05:30 mohammadkaif-ubuntu sudo: pam_unix(sudo:session): session closed for user root

2. The "Syslog" Section (System Integrity)

- **What it tracks:** General system activity, including service starts/stops, kernel alerts, and application errors.
- **Security Value:** This helps detect **Exploit attempts** that might cause a service to crash or behave unexpectedly.
- **Report Highlight:** With 3,714 events captured, you have proven that the SIEM is successfully monitoring the "heartbeat" of the Ubuntu target.



Time	Event
2/2/26 7:15:01 598 PM	2026-02-21T19:15:01.598482+05:30 mohammadkai-f-ubuntu CROK(9781): (root) CMD (command -v debian-sal > /dev/null && debian-sal 1 1)
2/2/26 7:12:36 765 PM	2026-02-21T19:12:36.765300+05:30 mohammadkai-f-ubuntu systemd[1]: Finished motd-news.service - Message of the Day.
2/2/26 7:12:36 764 PM	2026-02-21T19:12:36.764422+05:30 mohammadkai-f-ubuntu systemd[1]: motd-news.service: Deactivated successfully.
2/2/26 7:12:36 742 PM	2026-02-21T19:12:36.742159+05:30 mohammadkai-f-ubuntu systemd[1]: Starting motd-news.service - Message of the Day...
2/2/26 7:12:31 975 PM	2026-02-21T19:12:31.975035+05:30 mohammadkai-f-ubuntu splunk[9649]: 2026-02-21 19:12:31.974 +0530 splunkd started (build d749cb7ea65) pid=9649
2/2/26 7:12:31 653 PM	2026-02-21T19:12:31.653255+05:30 mohammadkai-f-ubuntu splunk[9649]: #01Validating installed files against hashes from '/opt/splunkforwarder/splunkforwarder-10.2.0-d749cb7ea65-linux-amd64-manifest'
2/2/26 7:12:31 652 PM	2026-02-21T19:12:31.652863+05:30 mohammadkai-f-ubuntu splunk[9649]: #01Checking default conf files for edits...
2/2/26 7:12:31 652 PM	2026-02-21T19:12:31.652657+05:30 mohammadkai-f-ubuntu splunk[9649]: #01Done
2/2/26 7:12:30 934 PM	2026-02-21T19:12:30.934348+05:30 mohammadkai-f-ubuntu splunk[9674]: New certs have been generated in '/opt/splunkforwarder/etc/auth'.
2/2/26 7:12:30 214 PM	2026-02-21T19:12:30.214298+05:30 mohammadkai-f-ubuntu splunk[9649]: #01Checking mgmt port [8883]: open

Windows

1. Methodology: Transitioning to Windows Monitoring

Document how you overcame the "Windows Home" limitations to ensure visibility:

- **The Challenge:** Windows Home Edition lacks the secpol.msc GUI for auditing, requiring manual configuration of the Universal Forwarder.
- **The Solution:** Manually created an inputs.conf file in C:\Program Files\SplunkUniversalForwarder\etc\system\local\ to force the ingestion of Security and System logs.
- **Permissions:** Confirmed the SplunkForwarder service was added to the **Event Log Readers** group to allow it to bypass Windows security restrictions.

2. Evidence: Verifying Data Flow

Show that your **DESKTOP-QKR30H3** machine is now a verified data source on Kali:

- **Internal Health:** Confirmed connection via index=_internal, showing health.log and splunkd.log.
- **Security Logs:** Verified the activation of the Windows Event Log monitor.

Failed Logins (Brute-Force Detection): index=* host="DESKTOP-QKR30H3" EventCode=4625

Purpose: Detects unauthorized attempts to guess passwords.

New Search

index=* host="DESKTOP-QKR30H3" EventCode=4625

Time range: Last 4 hours

403 events (2/21/26 3:26:00.000 PM to 2/21/26 7:26:31.000 PM) No Event Sampling

Events (403) Patterns Statistics Visualization

Timeline format Zoom Out Zoom to Selection Deselect

1 minute per column

Format Show: 20 Per Page View: List

Hide Fields All Fields

SELECTED FIELDS

- # host 1
- # source 1
- # sourcetype 1

INTERESTING FIELDS

- # Account_Domain 2
- # Account_Name 2
- # Authentication_Package 1
- # Caller_Process_ID 1
- # Caller_Process_Name 1
- # ComputerName 1
- # EventCode 1
- # EventType 1
- # Failure_Reason 1
- # index 1
- # Key_Length 1
- # Keywords 1
- # Inccount 1

#	Time	Event
1	2/21/26 5:15:49.376 PM	LogName=Security EventCode=4625 EventType=8 ComputerName=DESKTOP-QKR30H3 Show all 61 lines host = DESKTOP-QKR30H3 source = WinEventLog:Security sourcetype = WinEventLog:Security
2	2/21/26 5:15:49.376 PM	LogName=Security EventCode=4625 EventType=8 ComputerName=DESKTOP-QKR30H3 Show all 61 lines host = DESKTOP-QKR30H3 source = WinEventLog:Security sourcetype = WinEventLog:Security
3	2/21/26 5:15:49.376 PM	LogName=Security EventCode=4625 EventType=8 ComputerName=DESKTOP-QKR30H3 Show all 61 lines host = DESKTOP-QKR30H3 source = WinEventLog:Security sourcetype = WinEventLog:Security

Successful Logins (Access Monitoring): index=* host="DESKTOP-QKR30H3" EventCode=4624

Purpose: Confirms when someone successfully enters the system.

New Search

index=* host="DESKTOP-QKR30H3" EventCode=4624

Time range: Last 4 hours

5,346 events (2/21/26 3:28:00.000 PM to 2/21/26 7:28:26.000 PM) No Event Sampling

Events (5,346) Patterns Statistics Visualization

Timeline format Zoom Out Zoom to Selection Deselect

1 minute per column

Format Show: 20 Per Page View: List

Hide Fields All Fields

SELECTED FIELDS

- # host 1
- # source 1
- # sourcetype 1

INTERESTING FIELDS

- # Account_Domain 7
- # Account_Name 10
- # Authentication_Package 2
- # ComputerName 1
- # Elevated_Token 2
- # EventType 1
- # Impersonation_Level 2
- # index 1
- # Key_Length 1
- # Keywords 1
- # Inccount 1

#	Time	Event
1	2/21/26 6:23:54.266 PM	LogName=Security EventCode=4624 EventType=8 ComputerName=DESKTOP-QKR30H3 Show all 70 lines host = DESKTOP-QKR30H3 source = WinEventLog:Security sourcetype = WinEventLog:Security
2	2/21/26 6:23:54.266 PM	LogName=Security EventCode=4624 EventType=8 ComputerName=DESKTOP-QKR30H3 Show all 70 lines host = DESKTOP-QKR30H3 source = WinEventLog:Security sourcetype = WinEventLog:Security
3	2/21/26 6:23:54.266 PM	LogName=Security EventCode=4624 EventType=8 ComputerName=DESKTOP-QKR30H3 Show all 70 lines host = DESKTOP-QKR30H3 source = WinEventLog:Security sourcetype = WinEventLog:Security

Member Added to Administrators Group: index=* host="DESKTOP-QKR30H3"
EventCode=4732

Purpose: Detects privilege escalation, where a normal user is given "God Mode" powers.

New Search [Save As] [Create Table View] [Close]

index=* host="DESKTOP-QKR30H3" EventCode=4732 [Time range: Last 4 hours] [Search] [Smart Mode]

102 events (2/21/26 3:30:00.000 PM to 2/21/26 7:30:21.000 PM) [No Event Sampling]

Events (102) Patterns Statistics Visualization

[Timeline format] [Zoom Out] [Zoom to Selection] [Deselect] 1 minute per column

[Format] [Show: 20 Per Page] [View: List] [Prev] 1 2 3 4 5 6 [Next]

	Time	Event
>	2/21/26 4:47:33.300 PM	82/21/2026 04:47:33.300 PM LogName=Security EventCode=4732 EventType=0 ComputerName=DESKTOP-QKR30H3 Show all 30 lines host = DESKTOP-QKR30H3 source = WinEventLog:Security : sourcetype = WinEventLog:Security
>	2/21/26 4:47:33.300 PM	82/21/2026 04:47:33.300 PM LogName=Security EventCode=4732 EventType=0 ComputerName=DESKTOP-QKR30H3 Show all 30 lines host = DESKTOP-QKR30H3 source = WinEventLog:Security : sourcetype = WinEventLog:Security
>	2/21/26 4:47:33.300 PM	82/21/2026 04:47:33.300 PM LogName=Security EventCode=4732

SELECTED FIELDS
host 1
source 1
sourcetype 1

INTERESTING FIELDS
Account_Domain 2
Account_Name 3
ComputerName 1
EventCode 1
EventType 1
Group_Domain 1
Group_Name 2
index 1
Keywords 1
linecount 1
LogName 1
Logon_ID 2